

mass spectrometry (ICP-MS). SEM analysis was performed using a Hitachi S-4800 with an acceleration voltage of 10 kV, while TEM imaging was conducted with FEI Tecnai G2 F30 S-Twin operating at an acceleration voltage of 300 kV. High-angle annular dark-field scanning TEM (HAADF-STEM), selected area electron diffraction (SAED) pattern, and energy dispersive X-ray spectrometer (EDX) mapping were obtained using a TEM (JEOL, JEM-2100F HR) operated at an acceleration voltage of 200 kV. Crystal information of the PCO_{NA} was determined through XRD (RIGAKU, SmartLab) conducted in θ -2 θ scan mode with a Cu K α 1 incident beam. XPS (Thermo VG Scientific, K-Alpha) measurements provided insights into the chemical compositions, oxidation states and charge transfer evidence, with the C 1s peak at 284.8 eV serving as the reference for binding energy calibration. ICP-MS (Agilent, ICP-MS 7700S) was employed to analyze the chemical composition of PCO_{NA}, with experiments repeated at least five times to ensure accuracy.

Electrical conductivity measurements

The geometric factors associated with the nanowire-formed electrically conductive channels assured that their overall resistance was significantly larger than that of the electrode contact resistance. Thus, the electrical properties measured could be predominantly attributed to the PCO_{NA}. Electrical conductivity assessments of the PCO_{NA} were conducted using a two-probe electrode system. AC impedance spectroscopy was carried out using a Solartron 1255 HF frequency response analyzer and EG&G PAR potentiostat across a frequency range of 0.1 Hz to 100,000 Hz, within a temperature range of 450 – 650 °C (at intervals of 50 °C), under varying oxygen partial pressure (ranging from 0.1 to 1 atm). AC impedance spectroscopy was applied to ensure the ability to isolate bulk (high frequency contribution) from electrode (low frequency) contributions and in-plane conductivity measurements were made while lowering the temperature at each annealing temperature. All the impedance spectra were fit to an equivalent RC circuit in which the low frequency intercept of the high frequency semicircle corresponds to the resistance of the PCO_{NA} at each measurement temperature. The experimental setup involved placing the samples in an alumina tube, with a gold current collector linked to a gold wire, while gas mixtures of oxygen and nitrogen were regulated through digital mass flow controllers. In-plane conductivity was monitored by recording voltage measurements every 0.5 s until equilibrium was reached using AC measurements. Initial conductivities of PCO_{NA} without infiltration were evaluated to establish reference values.

Surface infiltration of binary oxides on PCO_{NA}

The infiltration method involved a simple drop and drying process on PCO_{NA} with ethanol-based solutions of tetraethylorthosilicate (TEOS) and Li(NO₃)₃. Solutions containing Li nitrate and TEOS at a concentration of 0.2 M were prepared for the infiltration process. Initial conductivities of PCO_{NA} without infiltration were evaluated to establish reference values. Infiltration was performed with the sample located in its original cell setup, ensuring complete solution coverage and drying of 50 μ L, thereby enhancing infiltration reliability. These precursors decompose and convert to binary oxides (SiO₂ and Li₂O, respectively) at temperatures exceeding 600 °C under air atmosphere²³. Thus, PCO_{NA} was annealed under air conditions at 650 °C and measurements initiated after stabilization of the conductivity was achieved.

Figures

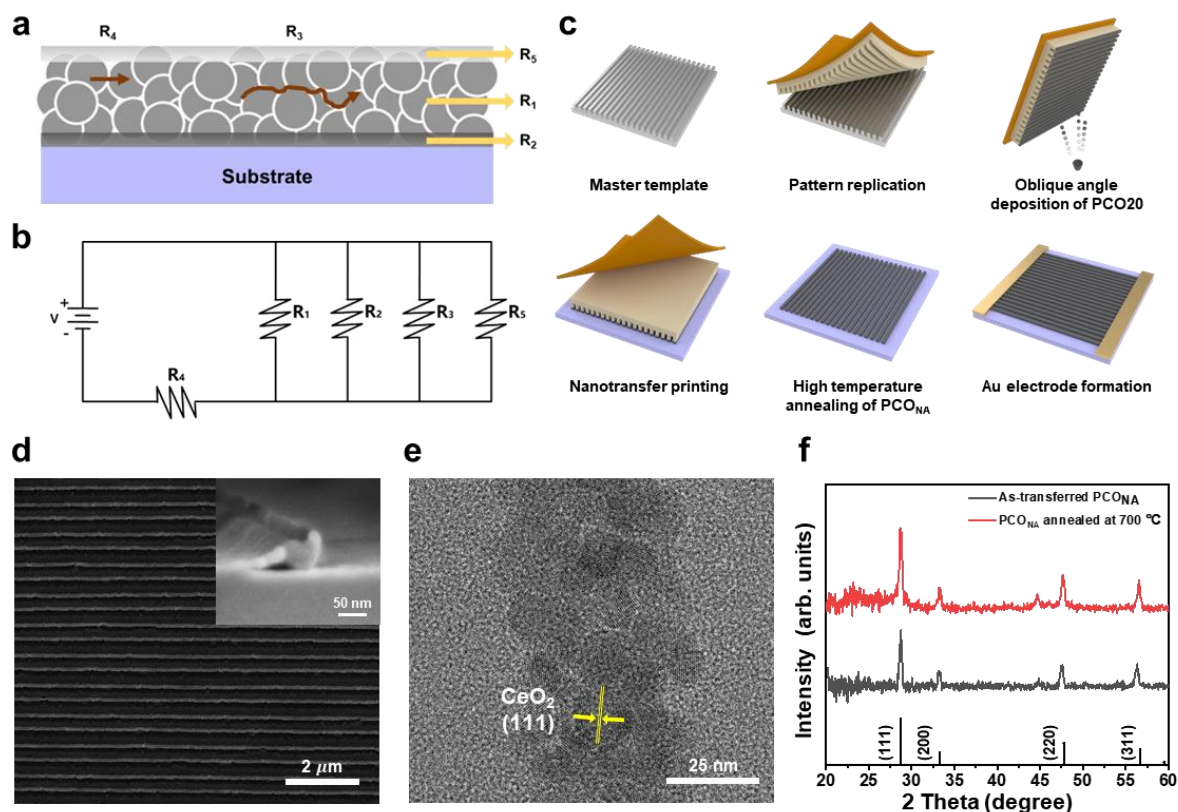


Fig. 1 Fabrication and characterization of PCO₂₀ nanowire arrays (PCO_{NA}) as model platform for analyzing effects of heterointerface acidity on electronic properties.

a Schematic cross-sectional illustration of several conduction pathways in PCO nanowires. **b** Corresponding equivalent circuit representing the interconnection of the conduction pathways illustrated in **a**. **c** Fabrication process of PCO_{NA} model platform through solvent-assisted nanotransfer printing method with pulsed laser deposition and high-temperature annealing. **d** SEM image of fabricated PCO_{NA} (inset image: Cross-sectional SEM image of fabricated PCO_{NA}) and **e** TEM image of fabricated PCO nanowire. **f** XRD spectra of as-transferred PCO_{NA} and fabricated PCO_{NA} after high-temperature annealing at 700 °C. Black vertical lines at bottom of figure indicate the dominant facets of the cerium oxide crystal structure.